

RENEWABLE ENERGY & ENVIRONMENTAL SUSTAINABILITY

Official Egyptian General Secondary Education Textbook (Grade 12)

ACADEMIC YEAR: 2024 / 2025 | GRADE 12 (THIRD SECONDARY)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Official Egyptian Ministry textbook for Renewable Energy & Environmental Sustainability: Solar PV & CSP engineering (Benban Solar Park), wind aerodynamics and Betz limit (Gabal El-Zeit & Zafarana), green hydrogen electrolysis and PtX corridors (SCZone), smart grid frequency droop, Ataka pumped hydro, and GHG Protocol / EU CBAM carbon border taxation.

CORE CURRICULUM HIGHLIGHTS:

- Photovoltaic I-V and P-V curve characterization, MPPT tracking, and NOCT thermal derating
- Wind kinetic cubic power formula, Betz limit $C_p=16/27$, and aerodynamic pitch regulation
- PEM vs Alkaline vs SOEC electrolysis thermodynamics, specific energy consumption, and green ammonia synthesis
- Unified grid frequency droop, BESS synthetic inertia, Ataka 2,400 MW pumped storage, and EU CBAM tariff liability

APPROVED CHAPTER MODULES:

- Solar Photovoltaic & Concentrated Solar Power (CSP) Engineering	pp. 1 68 (4 core units)
- Wind Turbine Aerodynamics & Betz Limit Dynamics	pp. 69 138 (4 core units)
- Green Hydrogen Electrolysis & Power-to-X (PtX) Integration	pp. 139 206 (4 core units)
- Smart Grids, BESS Energy Storage & Carbon Accounting	pp. 207 275 (4 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Renewable Energy, Clean Tech & Decarbonization Compendium (8p - 29.
Full Ministerial Textbook: 275 Pages - 32.6 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-RENEW-2025
Legal Deposit: 2025/14892 - ISBN: 978-977-12-1013-41



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Official Egyptian Ministry textbook for Renewable Energy & Environmental Sustainability: Solar PV & CSP engineering (Benban Solar Park), wind aerodynamics and Betz limit (Gabal El-Zeit & Zafarana), green hydrogen electrolysis and PtX corridors (SCZone), smart grid frequency droop, Ataka pumped hydro, and GHG Protocol / EU CBAM carbon border taxation. Photovoltaic I-V and P-V curve characterization, MPPT tracking, and NOCT thermal derating Wind kinetic cubic power formula, Betz limit $C_p=16/27$, and aerodynamic pitch regulation PEM vs Alkaline vs SOEC electrolysis thermodynamics, specific energy consumption, and green ammonia synthesis Unified grid frequency droop, BESS synthetic inertia, Ataka 2,400 MW pumped storage, and EU CBAM tariff liability

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
---	---	--

3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Solar Photovoltaic & Concentrated Solar Power (CSP) Engineering	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Wind Turbine Aerodynamics & Betz Limit Dynamics	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Green Hydrogen Electrolysis & Power-to-X (PtX) Integration	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Smart Grids, BESS Energy Storage & Carbon Accounting	7 wks	15 pts	25.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism)

Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.

Engineering Sciences (Civil, Mechanical, Electrical)

Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.

Computer Science & Algorithmics

Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: SOLAR PHOTOVOLTAIC & CONCENTRATED SOLAR POWER (CSP) ENGINEERING

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Solar Photovoltaic & Concentrated Solar Power (CSP) Engineering.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Solar Photovoltaic & Concentrated Solar Power (CSP) Engineering.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: WIND TURBINE AERODYNAMICS & BETZ LIMIT DYNAMICS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Wind Turbine Aerodynamics & Betz Limit Dynamics.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Wind Turbine Aerodynamics & Betz Limit Dynamics.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \sum (\omega_k * \psi_k) / \sum \omega_k$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{(1/N) * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: GREEN HYDROGEN ELECTROLYSIS & POWER-TO-X (PTX) INTEGRATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Green Hydrogen Electrolysis & Power-to-X (PtX) Integration.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Green Hydrogen Electrolysis & Power-to-X (PtX) Integration.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

RENEWABLE ENERGY & ENVIRONMENTAL SUSTAINABILITY - Core Principles	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$
Normalized Standard Score:	$Z = \frac{(X - \mu)}{\sigma}$
PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean:	$H = n / \left(\sum_{i=1}^n \frac{1}{x_i} \right)$
Sample Variance Boundary:	$s^2 = \frac{1}{(n - 1)} \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = \frac{\text{Output}_{\text{actual}}}{\text{Input}_{\text{standard}}}$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: TH-RENEW-2025
Curriculum Cycle: 2025/2026 Academic Year | National Edition
Official Portal: <https://moe.gov.eg/ar/elearning-content/>
All rights reserved (c) 2025 Ministry of Education & Technical Education - Egypt.
Unauthorized commercial reprinting or distribution without ministerial sanction is prohibited.